

FIR Intelligence & Crime Pattern Detector

Connecting the dots police can't see.

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3 crore FIRs. Zero connections.

3 crore+

digitised FIRs in UP Police's CCTNS, with
no NLP layer on top

Free text nobody can search by pattern.

1

Records, not insight

Every FIR is stored, but none is compared with the others.

2

Dots never connected

The Jamtara gang evaded detection for years because inter-district FIR links were never surfaced.

3

Analysis is manual

Officers must read FIRs one by one to spot a serial offender.

The information already exists. It just isn't connected, so repeat offenders look like first-time offenders.

An AI that reads FIRs like a detective

1

Categorise

Theft, burglary, cyber fraud, kidnapping and more

chain snatching

2

Extract

Accused, location, MO and victim profile

MO: poses as delivery agent

3

Detect

Flags FIRs likely tied to the same offender or gang

FIR A ↔ FIR B

4

Summarise

Station-level trend summary and a repeat-offender list

3-FIR series flagged

Example outputs, illustrative

A decision-support assistant for investigators, never an automatic accuser.

Five steps, one app



WHAT ONE FIR LOOKS LIKE GOING IN AND COMING OUT (ILLUSTRATIVE)

"Two men on a bike snatched a gold chain from a woman near the bus stand. One posed as a delivery agent asking for an address."



```
{ "crime_type": "chain snatching",  
  "accused_description": "two men, bike",  
  "location": "bus stand",  
  "mo_summary": "delivery-agent ruse, snatch",  
  "victim_profile": "woman, lone pedestrian" }
```

Everything runs from a single Python file: no separate backend, no separate frontend.

Where IBM Bob fits



IBM Bob

The reasoning engine behind two steps
of the pipeline

1

Reads each FIR

One combined prompt returns crime type, accused, location, MO and victim profile as JSON.

2

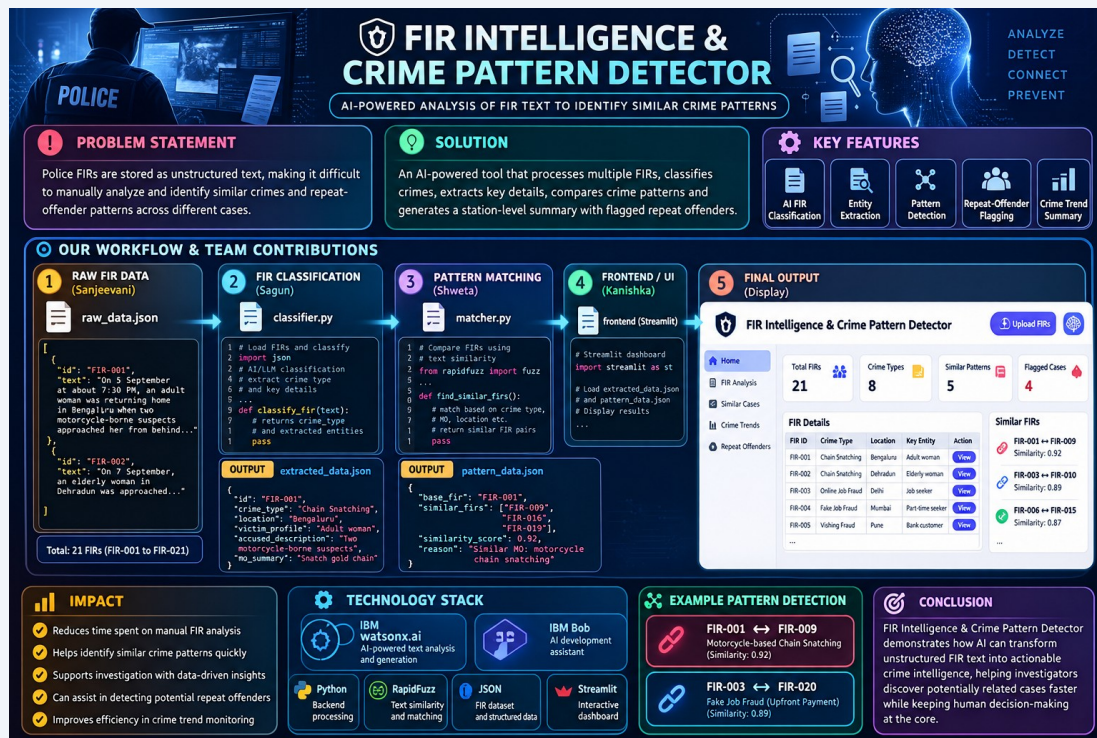
Writes the summary

Turns crime counts and flagged groups into one plain-English paragraph.

Swap-ready by design. Matching and the dashboard are plain Python, so the AI layer is a one-line endpoint change.

Bob handles language understanding; deterministic Python handles the matching, so results are explainable.

Watch it catch a hidden pattern



Matches on MO

not on names, numbers or dates

A plain-English reason

for every flag

Decoys stay out

FIR007, a lone thief in a market, is not linked

What success looks like

18

mock FIRs analysed

2

planted repeat-offender series to find

3

decoy FIRs that must stay unflagged

Scenario	Without the tool	With the tool
Serial chain snatcher	3+ months to link the cases	Flagged after the 3rd or 4th FIR
Fake-job cyber ring	A year of losses, six stations blind	One network flagged within days
Habitual burglar	Three unlinked "first offences"	Full pattern ready before bail

Demo-ready: one command (streamlit run app.py), no backend to start. Scenarios are illustrative; add measured results after the test run.

The team

1

Sanjeevani

Data

Built the mock FIR dataset with
planted series and decoys

2

Sagun Singh

Team Lead · AI prompts

Wrote the combined classify-
and-extract prompt

3

Shweta Rekha Nayak

Matching

Built the repeat-offender
matcher

4

Kanishka Mina

Streamlit app

Built the app that ties it all
together

Next: 60+ FIRs · LLM-written match rationales · downloadable PDF reports · station-level dashboards

All data is invented. A human reviews every flag. Thank you.